

SVD Applications

PCA (Principal Component Analysis)

This is a technique to analyze and learn information about large data sets. To see how linear algebra becomes involved, consider a data set with n samples and m measurable variables

	1	2	...	n
1	*	*	...	*
2	*	*	...	*
...	...	...	...	...
m	*	*	...	*

This can clearly be encoded as an $m \times n$ matrix A_0 . We now center the data by subtracting the means from each row and denote the resulting matrix A .

$$A_0 = [a_{ij}] \Rightarrow A = [a_{ij} - \mu_i], \quad \mu_i = \frac{1}{n} \sum_{j=1}^n a_{ij}$$

$$\begin{aligned} \text{Now consider } AA^T &= \begin{bmatrix} -\vec{A}_1 - \\ \vdots \\ -\vec{A}_m - \end{bmatrix} \begin{bmatrix} \vec{A}_1^T & \dots & \vec{A}_m^T \\ | & & | \end{bmatrix} \\ &= [\langle \vec{A}_k, \vec{A}_l \rangle]_{1 \leq k, l \leq m} \end{aligned}$$

The diagonal entries of AA^T are the sample variances $s_k^2 = \|\vec{A}_k\|^2$ and the off diagonal entries are the sample covariances $s_{kl} = \langle \vec{A}_k, \vec{A}_l \rangle$.

The covariance measures how the rows are correlated, high (positive) covariance means strong correlation. Low or negative covariance means low correlation or the variables are uncorrelated.

Dividing by the "degrees of freedom" in the sample we define the sample covariance matrix

$$S = \frac{AA^T}{n-1}$$

If units of measurement are a concern we use $n-1$

a diagonal matrix D s.t. DA has rows of length $\sqrt{n-1}$. Then we define the sample correlation matrix $C = \frac{DA A^T D^T}{n-1}$ which has ones along the diagonal.

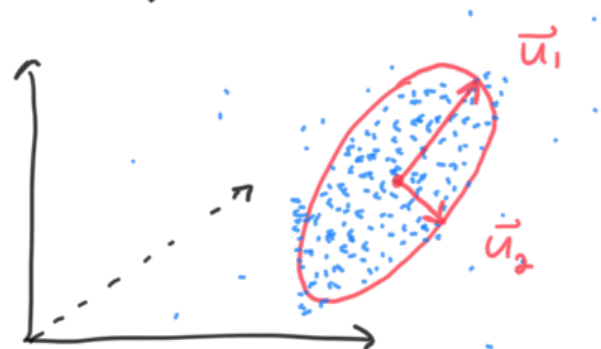
Crucial link: The total variance of the data is given by

$$T = \sum_{k=1}^m S_k^2, \text{ sum of the sample variances}$$

Now let σ_k^2 be the e-values of $S = \frac{A A^T}{n-1}$. We then have that

$$T = \sum S_k^2 = \sum \sigma_k^2$$

Thus the e-vectors of S corresponding the largest e-values represent the most significant directions in the data. Let $\sigma_1^2, \dots, \sigma_R^2$ be such that $\sigma_1^2 + \dots + \sigma_R^2 \sim T$ and let $\vec{u}_1, \dots, \vec{u}_R$ be the corresponding e-vectors of S . Then the data set is near the R -dim subspace with basis $\{\vec{u}_1, \dots, \vec{u}_R\}$. These vectors are the principal components of the data.



In a 2-dim setting, the principal component $\vec{u}_1$ is the orthogonal regression line and $\vec{u}_2$ represents the spread in the data.

Facial Recognition

Training set of n -faces $\rightarrow$ $m \times n$ matrix with each column representing the grayscale pixels of the face



(Nearly) Span the face space with the eigentfaces

find k most significant singular values and singular vectors, $\{\vec{v}_1, \dots, \vec{v}_k\}$
eigenfaces

And others !